Maths Ordinary Differential Equations

Engineering Technology

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Maths Ordinary Differential Equations (A11884)

Short Title:	Maths ODE		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Mathematics and Physics		
Author	CLAIRE.FITZPATRICK		
Credits	5	**Level:**	Introductory

Description of Module / Aims

This module gives an introduction to methods (analytical and numerical) for the solution of ordinary differential equations of first and second order together with necessary background material in multivariable differential calculus and complex variables.

Programmes

MTHE-0030	BEng (Hons) in Electrical Engineering (WD_ETRIC_B)	2 / 3 / M
	BEng (Hons) in Electrical and Automation Engineering (International) (WD_ETRIC_BI)	2 / 3 / M
MTHE-0030	BEng (Hons) in Electronic Engineering (WD_EONIC_B)	2 / 3 / M
	BEng (Hons) in Information Engineering (International) (WD_EEELC_BI)	2 / 3 / M
MTHE-0030	BEng (Hons) (WD_ETRIC_B)	2 / 3 / M
MTHE-0030	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	2 / 3 / M

Indicative Content

- Engineering Functions: Error function, Gamma function, Dirac delta function. Functions of a complex variable. Functions of two real variables; surface and contour plots. Partial Derivatives; Differentials. Fourier sums.
- Linear Ordinary Differential Equations (analytical methods): Existence, uniqueness and independence of solutions; Wronskian. First Order (integrating factor method, undetermined coefficient method). Second order (undetermined coefficients; variation of parameters). Solution of a system of linear first order equations using matrix exponential. Laplace transform methods.
- Nonlinear Ordinary Differential Equations (analytical and numerical methods). Separation of variables method; exact differential method; Euler and Runge-Kutta methods. Equilibrium solutions and stability.
- Qualitative methods for the analysis of the solutions of a pair of coupled ODEs

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Classify ordinary differential equations and hence determine whether a unique solution exists.
2. Review the general properties of an unseen Special Function using an Engineering handbook or a help file.
3. Recognise standard functions of two variables, find partial derivatives and differentials for such functions.
4. Use standard methods (analytical and numerical) to solve linear and basic nonlinear ordinary differential equations of first and second order and a system of first order linear equations.
5. Describe functions of a complex variable in terms of their real valued real and imaginary parts.
6. Use mathematical software to study ordinary differential equations.

Learning and Teaching Methods

- Delivery of the subject will be through a mixture of lectures and directed problem-solving sessions.
- The lectures will develop theory and present the associated mathematical methods.
- The directed work sessions will underpin and rehearse the skills demonstrated. Students will be required to construct valid and precise mathematical arguments and will be expected to start to treat mathematics as a modelling language rather than as a stand alone subject.
- The laboratory sessions consist of a number of computer based practicals to reinforce the concepts covered during the lectures using mathematical software such as OCTAVE/MATLAB or Python (or similar language).

Assessment Methods

	Weighing	**Outcomes Assessed**
Final Written Examination	70%	1,2,3,4,5,6
Continuous Assessment	30%	
Practical	15%	3
In-Class Assessment	15%	1,2,4,5,6

Assessment Criteria

<40%: Has not attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter or demonstrated a sufficient level of the skills expected on completion of this module.

40%-49%: Has attained the module learning outcomes at a basic level, demonstrated a basic knowledge of the associated subject matter, and achieved a basic level of the skills required for the subject matter.

50%-59%: Has attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

60%-69%: Has attained the module learning outcomes at a very good level, demonstrated a detailed knowledge of the associated subject matter, and achieved a very good level of the skills required for the subject matter.

70%-100%: Has attained the module learning outcomes at an excellent level, demonstrated a comprehensive knowledge of the associated subject matter, achieved an excellent level of the skills required for the subject matter, and demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

Learning Modes

Learning Mode	**F/T Hours**	**P/T Hours**
Lecture	48	
Tutorial		
Practical		
Independent Learning	87	

Supplementary Material

- Boyce, W.E. and R.C. DiPrima. _Elementary Differential Equations and Boundary Value Problems_. .USA: Wiley, .2012.
- Greenberg, M.D. _Advanced Engineering Mathematics_. USA: Prentice Hall, .2006.
- Nunez-Iglesias, J., S. van der Walt and H. Dashnow. _Elegant SciPy: The Art of Scientific Python_. USA: O'Reilly Media, 2017.

Requested Resources

- Room Type: Computer Lab